10 (a) Fig. 10.1 represents different positions A–H of the Moon as it rotates around the Earth.

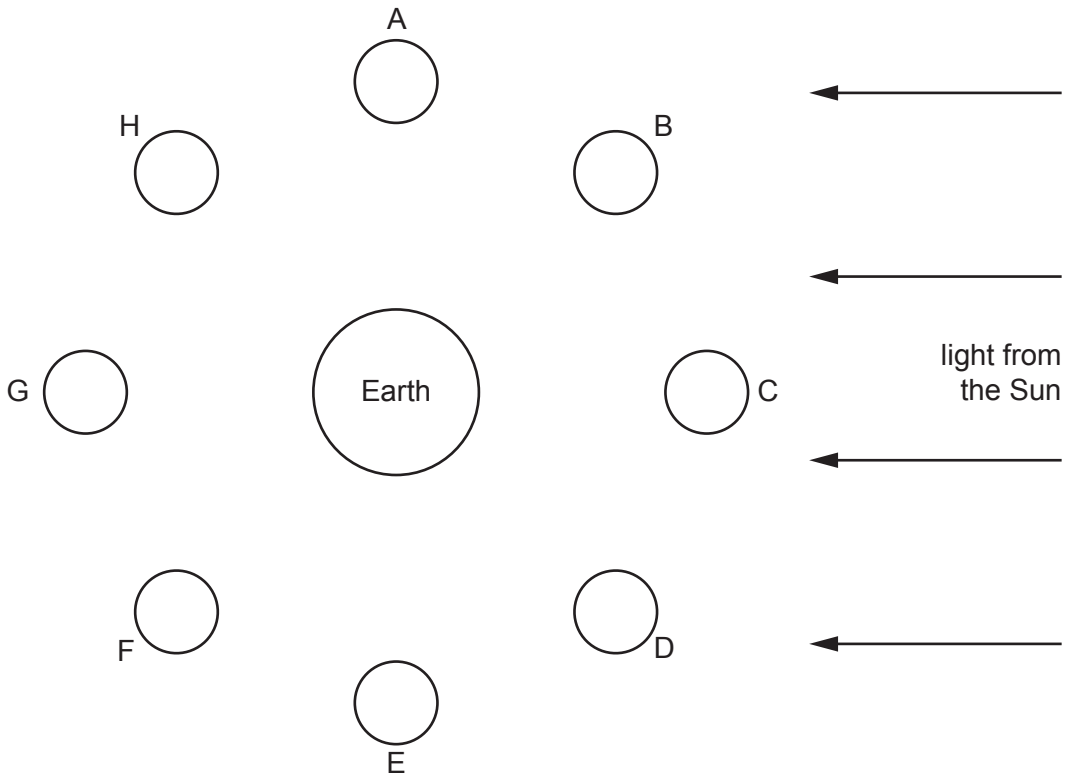


Fig. 10.1

(i) State a position of the Moon where an observer on Earth sees:

1. there is a quarter Moon  .....

2. there is a full Moon  .....

[2]

(ii) State the approximate time taken for the Moon to orbit the Earth.

time = ... [1]

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(b) The average distance of the Earth from the Sun is 1.5×10^8 km.

(i) Calculate the average orbital speed of the Earth in km / h.

average orbital speed = ... km / h [3]

(ii) The speed of light in a vacuum is 3.0×10^8 m / s. Calculate the time taken for light from the Sun to reach the Earth.

time = ... [2]

[Total: 8]